Design Process

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1. Demand analysis

Analysis captures the unprocessed, raw requirements from students, educators, and internal stakeholders during the project initiation stage. These are factual, unmodified descriptions of what the users want, serving as the foundational truth before we translate them into technical features.

Objectives:

1. Guidance and benchmark for product design Ensures that every feature developed directly serves the raw user need for fast, high-retention learning. It sets concrete benchmarks, such as a strict 90-second limit for content consumption per module.
2. Statement for product expectations and design boundaries Establishes clear boundaries for product designers. For example, the UX must remain mobile-first and swipe-based, preventing designers from veering into complex, desktop-heavy dashboard layouts.
3. Solution for common design problems (poor communication/lack of information) Acts as a single source of truth. When cross-functional teams disagree on feature prioritization (e.g., whether to build a social chat feature), the team reverts to the raw needs analysis to verify if users actually requested it.
4. Prevention against risks and mistakes (protection for designers) Prevents "feature creep" and protects designers from over-engineering the platform. By sticking to the raw need of *simplification*, we avoid the risk of building an overly complex LMS (Learning Management System).Description

Original Requirements Description:

- User Feedback (College Student): "Textbooks are exhausting. I spend hours scrolling TikTok anyway; I wish my biology notes were structured like my Reels feed so I could actually study while laying in bed or riding the bus."

- User Feedback (High School Student): "When I'm cramming the night before a test, I don't have time to re-read 50 pages. I just need the absolute core formulas and definitions highlighted, with a quick way to test myself."
- Stakeholder/Educator Request: "Students aren't reading the material. We need a way to deliver the absolute essentials of a lecture in under two minutes, ensuring they grasp the core concepts before coming to class."

Design Purpose

The purpose of this project is to bridge the gap between dense academic material and modern digital consumption habits. We are designing a platform that transforms heavy text into engaging, bite-sized, and mathematically accurate micro-content, maximizing study efficiency during daily downtime.

Usage Scenarios

- Scenario A (The Morning Commuter): * *Starting Point:* Sarah opens the app while boarding a crowded 15-minute subway ride to school.
    - *Situation:* She cannot comfortably open a textbook or laptop. She selects her "Organic Chemistry" module and uses her thumb to swipe through a vertical feed of 5 auto-generated flashcards and a 60-second audio summary, effectively studying a full chapter before her train arrives.
- Scenario B (The Midnight Crammer): * *Starting Point:* Alex is feeling overwhelmed at 11:00 PM by an upcoming history exam.

Situation: He uploads his professor's 30-page slide deck into the platform. The system instantly extracts the core timeline and definitions into an interactive "Cheat Sheet" feed, allowing him to grasp the big-picture concepts in under 5 minutes before

2. User Analysis

Target users for this platform are active learners who consume educational content in condensed, micro-dose formats. While institutional buyers (like universities or parents) may purchase subscriptions, the target users are the individuals interacting with the short-form content daily to retain knowledge efficiently.

Objectives

1. Make clear whom the product is designed for

Our product specifically targets time-constrained students and digital-native professionals who reject traditional, dense textbooks in favor of fast, high-retention learning. Design choices must prioritize rapid information delivery over long-form reading.

2. Mine the motivation behind user requirements

Users demand short-form study material because they experience cognitive overload, short attention spans, and time scarcity. Their core motivation is to achieve maximum academic or professional retention with minimal friction, turning passive "scrolling time" into active study sessions.

3. Extract design inspirations from user attributes

By analyzing user behavior, we unlock critical design directions:

- User Demand: The need to study during commutes $\rightarrow$ Design Inspiration: AI-generated audio bytes and bite-sized flashcards.
- Derived Function: High engagement with social media feeds

- Design Inspiration: A vertical-swipe "Edu-Feed" mimicking TikTok/Reels layout for micro-learning.

Description

Target User

The Time-Poor Micro-Learner: High school/college students cramming for exams, and young professionals upskilling on the go.

Basic Attributes

- Age: 16–30 years old (Gen Z and younger Millennials).
- Education Level: High school, university undergraduate, or entry-level corporate professionals.
- Tech-Savviness: Extremely high; primary device is a smartphone.
- Industry/Setting: Education, Tech, Business, or any fast-paced academic environment.

Characteristics & Attributes

Psychological Attributes:

- Attention Span: Low tolerance for walls of text; prone to digital distraction.
- Motivation Style: Extrinsically motivated by quick wins (streaks, badges) and intrinsically motivated by self-improvement.
- Cognitive Preference: Visual and auditory learner; prefers conceptual summaries over deep-dive historical context.

Behavioral Attributes:

- Screen Time: 4+ hours daily on mobile devices, heavily concentrated on short-form media apps.
- Study Habits: Prefers "micro-bursts" of studying (5–10 minute sessions during commutes, breaks, or right before bed) rather than 3-hour

library blocks.

- Sharing Tendency: Highly likely to screenshot and share interesting facts or concise explanations with peers via chat apps.

Design Inspiration

User Characteristic	Potential Function / Feature
Short attention spans	"The 60-Second Summary": Core concepts broken down into scrollable, bite-sized text cards or mini-infographics.
High mobile screen time	Gamified Daily Streaks: Push notifications that prompt a 2-minute daily review quiz to maintain a study streak.
Commuter habits	Text-to-Speech Playlists: An automated audio feed that reads out short-form study summaries like an educational podcast.
Exam stress	"Cheat Sheet" Generator: A feature allowing users to convert an entire uploaded PDF chapter into 5 core summary bullet points instantly.

<table>
<tr><td>Target User</td><td colspan="2">The "Micro-Dose" Learner: High school/university students and fast-paced young professionals who struggle with traditional learning methods</td></tr>
<tr><td colspan="2">Q: What is your biggest frustration when attempting to review heavy academic text or study guides?
A: "Sitting down with a dense, 40-page chapter feels completely overwhelming. My brain shuts off after ten minutes. It takes too much mental energy just to figure out what the actual core takeaways are."</td><td>Q: How do you typically behave during small gaps in your day, like commuting or waiting in line?
A: "I mindlessly scroll through vertical video feeds on my phone. I feel guilty about wasting time. If my study materials were delivered in the exact same, visual format, I would swipe through them to study."</td></tr>
<tr><td colspan="3">User Portrait</td></tr>
<tr><td>Attributes</td><td>Attribute Description</td><td>Design Inspirations</td></tr>
<tr><td rowspan="2">Basic Attributes</td><td>Age: 16–25 (Gen Z students).</td><td>The entire application interface layout must prioritize one-handed thumb navigation.</td></tr>
<tr><td>Smartphone-exclusive for casual consumption.</td><td>Allow users to download short study packets for uninterrupted reading underground or during commutes.</td></tr>
<tr><td>Characteristics Attributes</td><td>8-second filter window; highly prone to digital distraction.</td><td></td></tr>
</table>

3. Stakeholder Analysis

Stakeholders for BrainNourishment encompass all individuals, groups, or systems that influence or are influenced by the platform's market adoption, content ecosystem, user experience, and overall commercial success. This includes not just the end-user (the student), but also the content suppliers, financial backers, and the technical development team.

1. Grow the app and the design over time in a direction Balancing stakeholder need. For example, ensuring content satisfies educational standards (educator requirement) prevents UX from becoming too entertainment-focused at the cost of academic accuracy.
2. Refine more comprehensive functional requirements and design inspiration Evaluating non-user stakeholders uncovers hidden features. Acknowledging that parents or institutions pay for the app inspires dashboard features like a "Weekly Progress Summary" email, which wouldn't be captured by studying the student alone.
3. Clarify the priority of demands When conflicts arise—such as a student wanting zero monetization friction versus an investor requiring ad revenue—this analysis acts as a framework to prioritize features for the Minimum Viable Product (MVP).

Stakeholder examples:

Stakeholder 1: Academic Educators & Content Creators

- Expectations/Requirements: * High fidelity and accuracy in the AI summarization pipeline; core academic concepts must not be diluted or misrepresented.
    - Easy-to-use ingestion tools to upload raw lecture notes, PDFs, or syllabus materials and review the short-form output before students see it.
- Objectives/Motivations: * Professional Value: To increase student engagement and conceptual preparation before lectures without increasing grading workloads.
    - Emotional Value: Pride in seeing students succeed and reduction in the frustration of teaching an unprepared class.

Stakeholder 2: Premium Buyers (Parents & Academic Institutions)

- Expectations/Requirements: * Tangible proof of value, such as clear, data-driven analytics demonstrating improved retention rates or test scores.
    - Robust data privacy compliance (such as COPPA/GDPR alignment) and safe, distraction-free ad environments.
- Objectives/Motivations: * Financial/Value Motivation: Peace of mind knowing that screen time is being utilized productively ("nourishing the brain") rather than mindlessly wasted on purely social algorithms.

Stakeholder 3: Internal Product & Engineering Team

- Expectations/Requirements: * Scalable infrastructure capable of processing dense, multi-page textbooks into short-form multimedia layouts (text, flashcards, audio) via automated pipelines.
    - High user-retention metrics and daily active user (DAU) loops to satisfy investors during funding rounds.
- Objectives/Motivations: * Operational Motivation: To build a technically stable, highly automated product with minimal manual content curation overhead, establishing BrainNourishment as a leading, scalable pioneer in the micro-learning EdTech space.

Stakeholder Analysis Worksheet

Stakeholder	Type	Expectations/Requirements	Objectives/Motivations	Design Inspiration	Remarks
The Student / Learner	Primary / External	• Zero-friction UI. • Content readable/audible in <2 mins. • Smooth swipe mechanics.	• Pass exams with minimal study effort. • Reduce cognitive overload. • Turn mindless scrolling into guilt-free productivity.	• Vertical "Edu-Feed" layout (TikTok style). • Interactive 10-second recall quizzes. • Haptic feedback on daily streaks.	Top priority. If the UI feels like a textbook or a clunky LMS, they will drop the app instantly.
Educators / Teachers	Secondary / External	• High factual accuracy from AI summaries. • Zero dilution of formulas/theorems. • Easy bulk uploading tools.	• Ensure students actually review core material before lectures. • Boost class performance without extra grading workload.	• "Verify & Publish" Portal: A web dashboard where teachers quickly approve AI-generated micro-content before push notifications go out.	Critical for institutional adoption and credibility. They provide the vetted source materials.
Parents / Buyers	Economic / External	• Tangible proof of learning (ROI). • Strict data privacy compliance. • Distraction-free interface.	• Redirect negative screen time into productive habits. • Digital devices are being used for self-improvement.	• Weekly Progress Digest: Auto-generated email summaries showing "Top concepts mastered" and "Total focus minutes."	They hold the credit card for premium tiers. The app must look and feel safe and constructive.

4. Competitor Analysis

Definition

Competitor analysis for BrainNourishment applies comparative analysis methodologies to study user behavior across existing platforms. By examining products that students currently use to study or pass the time, we systematically break down their UX patterns, feature sets, and pitfalls to define where BrainNourishment can capture uncontested market share.

Objectives

1. Quick product modeling
    - Framework reference: Accelerates UI layout definition by adopting widely accepted mobile standards (e.g., standard media-player buttons, card swipe mechanics).
    - Opponent monitoring: Keeps track of how legacy EdTech platforms adapt to shortening user attention spans.
    - Quick trial and error: Allows us to skip features that competitors spent capital testing but ultimately discarded due to poor user retention.
2. Help design innovation
    - Identifies the massive market gap between high-engagement social entertainment (TikTok) and low-engagement academic utility (Quizlet).
    - Provides direct inspiration on how to marry algorithmic content delivery with scientifically proven learning frameworks (like active

recall and spaced repetition).

Description

- Step One: Competitor Identification: We gathered data across three core categories: legacy flashcard applications, content summarization platforms, and short-form social video feeds.
- Step Two: Competitor Selection: Selected three distinct, high-market-validation products that each own a piece of our intended user behavior: Quizlet, Blinkist, and TikTok.
- Step Three: Competitor Breakdown: Segmented these apps into core behavioral layers: how they present information, how users navigate content, and how they drive daily user retention.
- Step Four: Function Integration: Synthesized their best elements into a specialized features list for BrainNourishment, giving top priority to AI text-chunking and vertical thumb-swiping mechanics.

Competitor analysis Worksheet

Competitor Name	Competitor Introduction	Core Function Points and Design Significance	Design Inspiration
Quizlet	Legacy EdTech platform focused on user-generated digital flashcards, active recall study modes, and basic AI practice tests.	• Digital Flashcards: Allows rapid review of terms. *Significance:* Proves students rely heavily on self-testing and bite-sized QA text. • Smart Grading AI: Automatically assesses open-ended answers. *Significance:* Lowers user friction during self-assessment modules.	• Automated Question Generation: Copy the backend logic that converts a static definition into a multiple-choice question instantly. • Horizontal Flips: Retain the tap-to-flip flashcard mechanic, but pivot the overall discovery feed to a vertical orientation.
Blinkist	Subscription-based text and audio summary app that condenses 300-page non-fiction books into 15-minute structured text/audio insights ("Blinks").	• Dual-Format Player: Seamless switching between reading text and listening to audio narration. *Significance:* Essential for on-the-go utility during commutes. • Categorized Key Takeaways: Breaks book summaries into numbered "chapters of insight." *Significance:* Reduces reading fatigue and creates natural stopping points.	• The "Audio-Study" - A prominent button allowing a student to turn chapters flashcard deck into a AI-voiced study. • Visual Text Anchors: Bold typography emphasizing critical academic concepts, definitions, and equations within the summary view.

5. Scenario Analysis

Definition

Scenario Analysis for BrainNourishment details the exact real-world environments, physical constraints, and emotional states of students using the app. By mapping these specific moments, we directly translate user behaviors into mandatory app features, layout constraints, and navigation rules.

Objectives

1. Deeply dig into the essential needs of users and carry out targeted designs Ensures short-form content scales dynamically to the user's physical environment, such as designing one-handed UI layouts for students in cramped spaces.
2. Switch to different angles and make diversified designs Triggers a shift from standard text-based UI to alternative accessibility layers, such as high-quality text-to-speech audio for eyes-free learning.
3. Think in others' shoes, especially users' Empathizes with the cognitive exhaustion, sleep deprivation, and extreme exam anxiety of the target user, keeping the interaction barrier as low as possible.
4. Find user pain points and excitement to improve product experience Isolates the exact friction points where a student gets overwhelmed and quits, replacing those moments with instant AI summaries and gamified rewards to build a daily study habit.

Mapped Scenarios & Functional Requirements

Scenario A: The Transit Micro-Learning Session

- Context: Leyla (20, college sophomore) is standing on a loud, bumpy city bus at 08:15 AM, trying to review 20 core psychology terms right before a midterm.
- Refined Functions & Rules: * The entire core interface must be fully operable using a single thumb (Thumb-Zone UI rule).
    - Text content must feature a one-tap toggle to switch instantly into an AI-voiced audio stream for headphone users.
    - Typography must include a quick-access macro font scaler to prevent eye strain caused by vehicle vibrations.

Scenario B: The Midnight Cram Session

- Context: Can (17, high school senior) is lying in bed at 11:45 PM panicking over a dense, 50-page history textbook chapter due the next morning. He is physically exhausted and paralyzed by the volume of text.
- Refined Functions & Rules:
    - The app must feature a low-friction drag-and-drop file ingestion tool that processes heavy PDFs into 5 core summary cards in under 5 seconds.
    - The UI must support a true OLED black dark mode to prevent eye strain in low-light environments.
    - Information must use progressive disclosure—showing the 3 absolute core facts first, with optional tap-to-expand details to prevent cognitive overload.

Scenario Analysis Worksheet

Scenario Post-it Notes	Scenario Analysis	Design Inspiration
Scenario 1 • **Time:** 08:15 AM (Morning rush) • **Location:** Crowded, bumpy city bus • **Characters:** Leyla (20, college student) • **Event:** Reviewing psychology definitions right before a midterm exam. *(Scenario 1 Continued)*	**Pain Points:** • Only has one free hand because she is holding a passenger safety rail. • Shaky bus motion causes immediate eye strain when reading text. **Advantages / Excitement:** • High baseline focus driven by immediate test urgency. • Productively captures "dead time" usually wasted on mindless social feeds.	• **Thumb-Zone UI Layout:** Position all primary interactive buttons (swipe, save, flip) exclusively at the bottom of the screen. • **One-Tap Audio Stream:** A prominent button to instantly route an AI-voiced audio summary to the user's headphones. • **Dynamic Font Scaler:** Quick gesture controls to enlarge text size on the fly. • **"High-Yield Mode" Filter:** A high-contrast toggle that hides secondary text, showing only core definitions and formulas for rapid scanning.
Scenario 2 • **Time:** 11:45 PM (Night before final) • **Location:** Dark bedroom desk / bed • **Characters:** Can (17, high school senior) • **Event:** Paralyzed and cramming a dense, 50-page history chapter.	**Pain Points:** • Severe cognitive exhaustion and high text anxiety. • Overwhelmed by textbook volume; high risk of giving up.	• **File-to-Card Parsing Ingestion:** A clean upload portal where users drop a heavy document and receive structured summary decks instantly. • **True OLED Dark Mode:** An ultra-low brightness, high-contrast visual theme optimized for dark rooms.

6. Function List

Definition

The Function List for BrainNourishment is the definitive catalog of feature specifications designed to meet the micro-learning requirements of our users. This list categorizes features into modules, assigns an execution priority rank, and establishes strict logic rules to ensure the product remains optimized for high-retention, short-form digital learning.

Objectives

1. Get a comprehensive list of alternative solutions Provides the product team with a diverse map of potential feature variations (such as audio-only playlists vs. visual swipe-cards) to choose the absolute leanest path to product-market fit.
2. Select the most suitable combination of functions Protects the application from feature creep by filtering out complex, unnecessary components (like full essay editors or peer-to-peer chat systems) that detract from the core premise of bite-sized information.
3. Provide guidance for upcoming product design cycles Serves as the foundational reference blueprint for UI/UX wireframing, technical database architecture, and QA verification testing during the engineering phase

Module	Function Name	Importance Level	Function Notes & Operational Rules
AI Content Ingestion	Smart Document-to-Card Parser	4 (Must have)	• Accepts PDF, DOCX, and text file uploads up to 50MB. • Backend automatically parses text into independent semantic chunks. • Enforces a strict limit of 150 words maximum per generated card.
AI Content Ingestion	Automated QA Generator	3 (Suggest having)	• AI scans the summary text and auto-creates 3 multiple-choice testing questions for active recall at the end of every deck module.
Core UI/UX Feed	Vertical Scroll "Edu-Feed"	4 (Must have)	• Core homepage navigation interface utilizing vertical swiping mechanics (matching short-form social media feeds). • Swiping up instantly caches and displays the next logical concept card.
Core UI/UX Feed	Tap-to-Flip Active Recall	4 (Must have)	• Tapping the active card performs a 3D flip animation to reveal the core definition, formula answer, or explanatory diagram on the back.
Audio Integration	Natural AI Text-to-Speech	3 (Suggest having)	• Converts visible card text into automated, natural-sounding audio. • Includes a persistent background background play toggle for continuous listening when the phone screen is locked.

7. Prototype Design

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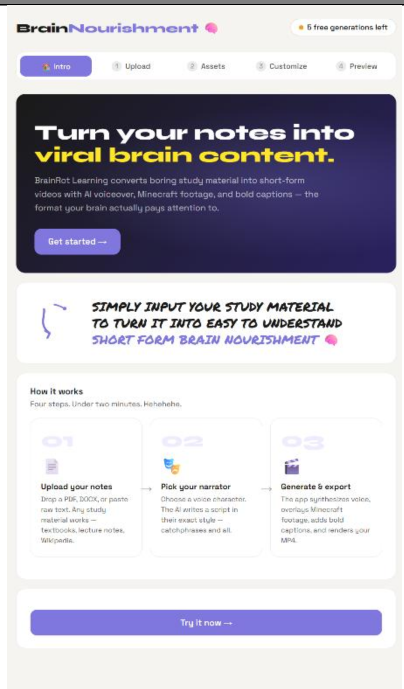
<tr><th colspan="2"></th></tr>
<tr><td>


</td><td>灵感|INSPIRATION

Inspiration is drawn from modern UI apps that are easy to use and easy to understand, suitable for a low-concentration users app</td></tr>
<tr><td colspan="2">总结|SUMMARY

The web user interface design features a clean, modern layout utilizing a light cream background contrasted with rounded container cards and a vibrant purple accent color for primary interactive elements. At the top, a persistent header balances a stylized logo on the left with a rounded</td></tr>
</table>

usage tracker on the right, sitting directly above a horizontal five-stage stepper that uses clean typography and numbered states to map out the user path. The main hero section uses a striking dark purple gradient card with prominent, bold white and yellow headline text to establish an immediate visual hierarchy, anchored by a rounded purple call-to-action button. Below this, a secondary white banner incorporates a playful, hand-drawn blue arrow graphic to direct the eye toward a stylized, uppercase script font. The informational section is organized into a clean three-column grid, where each card displays a large faded number indicator, a simple icon, a bold subhead, and compact descriptive text, all linked sequentially by subtle transition arrows. The entire page design finishes with a highly visible, full-width purple call-to-action button at the bottom to anchor the layout and drive user interaction.